

# Physics 2

## Comprehensive Review Package

Professor-first master review and reusable knowledge base

Point-charge force and field



This package consolidates the saved Canvas pages and worksheet scans into a conceptual progression. Each section gives the main ideas, a hallmark equation, the physical reasoning behind its derivation, misconceptions, and an immediate conceptual check with answer. Textbook supplements are labeled in the source map.

# Contents and coverage map

01. Charge and conservation
02. Coulomb force and electric fields
03. Field lines, superposition, and vectors
04. Continuous charge distributions and integration
05. Electric flux and Gauss law
06. Applications of Gauss law
07. Conductors, shielding, and induced charge
08. Electric potential, energy, and equipotentials
09. Capacitance and dielectrics
10. Dipoles, torque, polarization, and displacement
11. Oscillations and simple harmonic motion
12. Mechanical waves, interactions, and standing waves
13. Sound, intensity, spectra, hearing, and Doppler effect

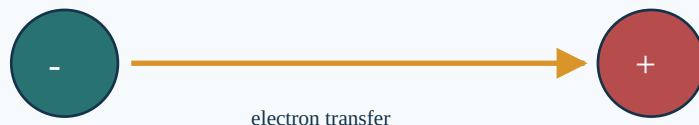
The companion YAML records source pages, equations in normalized LaTeX, assumptions, limiting cases, diagrams, checks, and question-generation tags. The page-list-only gaps identified during inventory are covered as supplemental sections where a saved capture was absent.

# 01. Charge and conservation

**Scope and source status:** Charge types, quantization, conservation, friction, contact, induction, conductors/insulators, electrostatics versus electrodynamics.

**Question-generation tags:** fill-in explain-phenomenon qualitative-prediction

Electron transfer and conservation



## Core understanding

- Charge is a signed, quantized property carried mainly by protons and electrons. A neutral object has equal positive and negative charge, not an absence of charge.
- Charging by friction or contact transfers electrons; charging by induction separates charge inside an object and requires grounding to leave a net charge.
- Conservation applies to the complete isolated system: one object gaining electrons means another loses the same electrons. Electrostatics describes settled charges; electrodynamics describes charge in motion or changing fields.

### Hallmark equation

$$Q_{\text{total}} = \text{constant}$$

**Symbols:**  $Q_{\text{total}}$  is the algebraic sum of all positive and negative charge in the chosen isolated system. **Physical meaning:** the equation says electrons may move between objects, but the system cannot gain or lose net charge. A local charge change is balanced by an equal and opposite change elsewhere.

**Concept check:** An initially neutral object gains electrons from a second object. Describe the final signs of both objects and explain how the total charge of the complete system changes.

**Answer and explanation:** An isolated object that gains electrons becomes more negative. The charge of the object changes, but the charge of the complete isolated system does not: electrons have moved from one part of the system to another. Induction is therefore a redistribution of existing charge, not creation of new charge.

## 02. Coulomb force and electric fields

**Scope and source status:** Inverse-square force, source/test charge roles, field direction, units, gravity analogy, and force from a field.

**Question-generation tags:** fill-in explain-phenomenon derivation

Point-charge force and field



### Core understanding

- Coulomb law gives the force between point charges. The force acts along the line joining the charges, attracts unlike signs, repels like signs, and falls as  $1/r^2$ .
- Electric field is force per unit positive test charge. It belongs to the source arrangement, while the actual force on a charge is  $F=qE$  and reverses direction for  $q<0$ .
- The gravity analogy is useful for the inverse-square geometry, but electric charge has two signs and can produce either attraction or repulsion.

#### Hallmark equation

$$F = q_1 q_2 / (4\pi\epsilon_0 r^2) \quad E = F/q_{\text{test}}$$

**Symbols:**  $q_1, q_2$  are Source charges,  $r$  is their separation,  $\epsilon_0$  is the vacuum permittivity, and  $\mathbf{r}_{\text{hat}}$  gives direction.  $E$  is force per unit positive test charge. **Physical meaning:** inverse-square spreading weakens interaction with distance; the signs determine attraction or repulsion, while  $E$  describes the source before a test charge is added.

**Concept check:** A negative test charge is placed at a point where the electric field points east. Predict the force direction and explain why field direction is defined with a positive test charge.

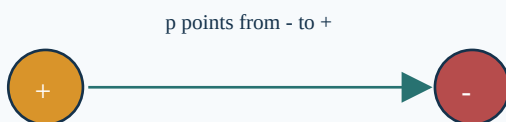
**Answer and explanation:** The force is opposite the field because  $F=qE$  and  $q$  is negative. The field direction is defined using a positive test charge, so a negative charge reverses that direction. This is why field and force must not be treated as interchangeable vectors.

### 03. Field lines, superposition, and vectors

**Scope and source status:** Vector addition, components, symmetry cancellation, field-line density and direction, point-charge limits, dipole far-field behavior.

**Question-generation tags:** diagram-matching draw-phenomenon graph-interpretation

Vector superposition and dipole



#### Core understanding

- Superposition means calculate each source field at the observation point and add the vectors. Components cancel or reinforce according to geometry; magnitudes alone are not enough.
- Field lines are a drawing convention: the tangent gives the direction for a positive test charge, and line density represents relative strength. Lines never cross.
- Dipole fields show cancellation clearly: far away, the fields of  $+q$  and  $-q$  nearly cancel, so the remaining field decreases faster with distance than a single-charge field.

#### Hallmark equation

$$\mathbf{E}_{\text{net}} = \sum_i \mathbf{E}_i$$

**Symbols:**  $\mathbf{E}_i$  is the field from source  $i$  and  $\mathbf{E}_{\text{net}}$  is the vector total. **Physical meaning:** each source acts independently, then the components combine. This equation explains cancellation on symmetry axes and reinforcement between opposite charges.

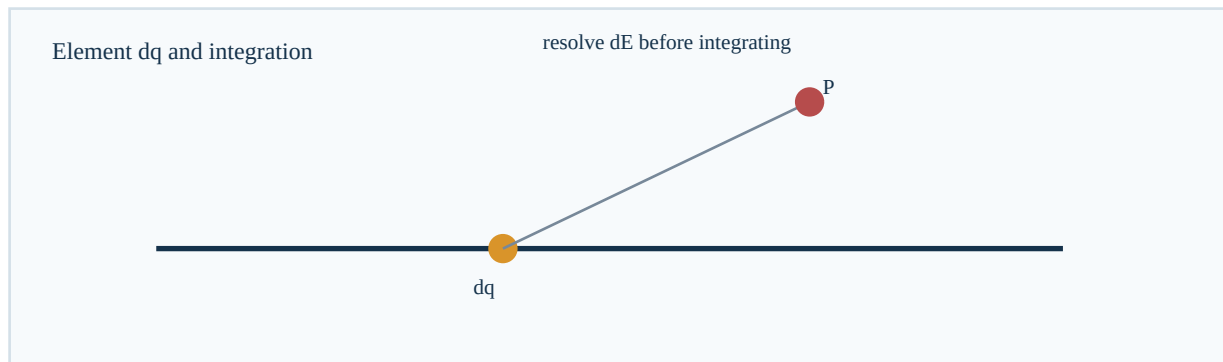
**Concept check:** Two equal positive charges are placed symmetrically about a point. Explain which field components cancel at the midpoint, and describe what changes if the charges have opposite signs.

**Answer and explanation:** Nonzero contributions can sum to zero when equal vectors point in opposite directions. Field-line density represents relative magnitude, while the tangent gives the direction a positive test charge would move. Dipole cancellation is why its distant field falls faster than a single-charge field.

## 04. Continuous charge distributions and integration

**Scope and source status:** lambda, sigma, rho, dq elements, total charge, line/arc/ring/disk/sheet/cylinder geometry, trigonometric substitutions, differential reasoning.

**Question-generation tags:** derivation draw-phenomenon fill-in



### Core understanding

- A continuous charge distribution is handled by dividing the source into elements  $dq$ . Use  $dq = \lambda dl$ ,  $\sigma dA$ , or  $\rho dV$  according to whether charge lies on a line, surface, or volume.
- For each element, write its distance and direction to the field point, resolve the vector into components, and integrate over the actual source limits.
- Symmetry is part of the solution: paired elements can cancel one component before integration. Trigonometric substitutions simplify the distance factor, but never replace the geometry or bounds.

#### Hallmark equation

$$dq = \lambda dl = \sigma dA = \rho dV \quad d\mathbf{E} = (1/4\pi\epsilon_0) dq/r^2 \hat{\mathbf{r}}$$

**Symbols:**  $\lambda$ ,  $\sigma$ , and  $\rho$  are line, surface, and volume charge densities;  $dl$ ,  $dA$ , and  $dV$  are source elements;  $dq$  is the charge in one element. **Physical meaning:** integration is a limit of adding many small Coulomb contributions whose distances and directions vary across an extended source.

**Concept check:** For a uniformly charged rod observed from its perpendicular bisector, explain how you would choose  $dq$ , which field component cancels, and why the remaining component must be integrated.

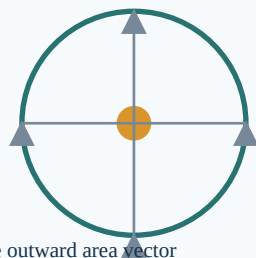
**Answer and explanation:** For a uniform rod,  $dq = \lambda dx$ . In a symmetric ring or rod arrangement, each element has a partner whose transverse component cancels. The remaining component is integrated after expressing  $r$  and the angle in terms of the chosen coordinate.

## 05. Electric flux and Gauss law

**Scope and source status:** Flux, area vectors, enclosed charge, closed surfaces, spherical/cylindrical/planar symmetry, and external-charge cancellation.

**Question-generation tags:** diagram-matching explain-phenomenon derivation

Closed Gaussian surface



### Core understanding

- Flux is the field component normal to a surface multiplied by area, summed over the surface. The area vector points outward on a closed surface.
- Gauss law says net closed-surface flux measures enclosed charge. External charges can change the field at individual patches but add zero net flux because their field enters and leaves the surface.
- Gauss law becomes a practical field solver only when symmetry makes  $E$  constant on a chosen Gaussian surface or makes parts of the flux zero.

#### Hallmark equation

$$\Phi_E = \oint \mathbf{E} \cdot d\mathbf{A} = Q_{\text{enc}} / \epsilon_0$$

**Symbols:**  $d\mathbf{A}$  is an outward area vector,  $E$  is the local field,  $Q_{\text{enc}}$  is net enclosed charge, and  $\epsilon_0$  is vacuum permittivity.

**Physical meaning:** the total excess of field passing outward through any closed surface is set only by charge inside it.

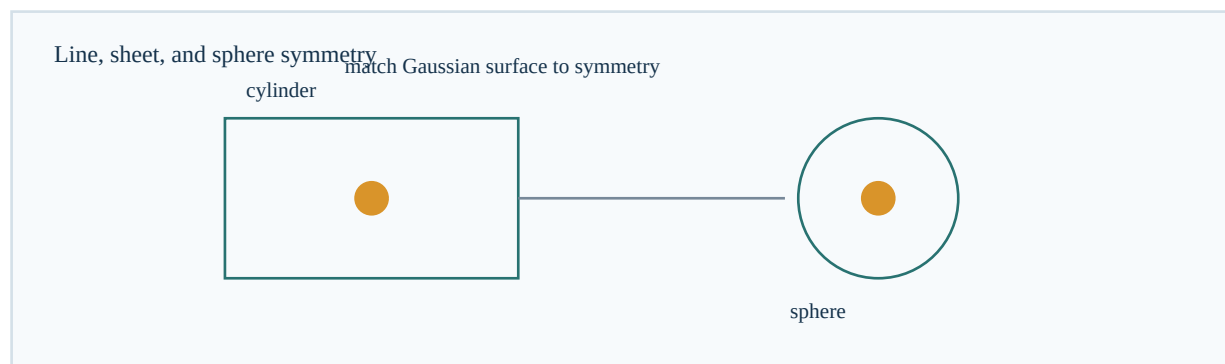
**Concept check:** A closed Gaussian surface encloses zero net charge but passes through a nonzero electric field. Explain how the net flux can still be zero.

**Answer and explanation:** The net flux is zero when the enclosed net charge is zero, even if the field is nonzero at individual points. External field lines enter and leave the closed surface, so their inward and outward contributions cancel in the total flux.

## 06. Applications of Gauss law

**Scope and source status:** Infinite line, sheet, sphere, nonuniform spherical charge, conductor surface, inside/outside field graphs.

**Question-generation tags:** derivation graph-interpretation qualitative-prediction



### Core understanding

- For an infinite line, cylindrical area grows as  $r$ , giving  $E$  proportional to  $1/r$ . For a sphere, area grows as  $r^2$ , giving the familiar inverse-square outside field.
- Inside a uniformly charged solid sphere, enclosed charge grows as  $r^3$  while Gaussian area grows as  $r^2$ , so  $E$  grows linearly with  $r$ .
- The same procedure works for sheets, shells, and conductors: determine enclosed charge first, then use the surface on which symmetry controls the flux.

#### Hallmark equation

$$E_{\text{line}} = \lambda / (2\pi\epsilon_0 r) \quad E_{\text{sheet}} = \sigma / (2\epsilon_0)$$

**Symbols:**  $\lambda$  is line charge density,  $\sigma$  is sheet density,  $r$  is distance from the symmetry axis or center, and  $\rho$  is volume density. **Physical meaning:** the field falloff follows how the available flux area grows: cylindrical area gives  $1/r$ , spherical area gives  $1/r^2$ , and a plane gives constant area per unit length.

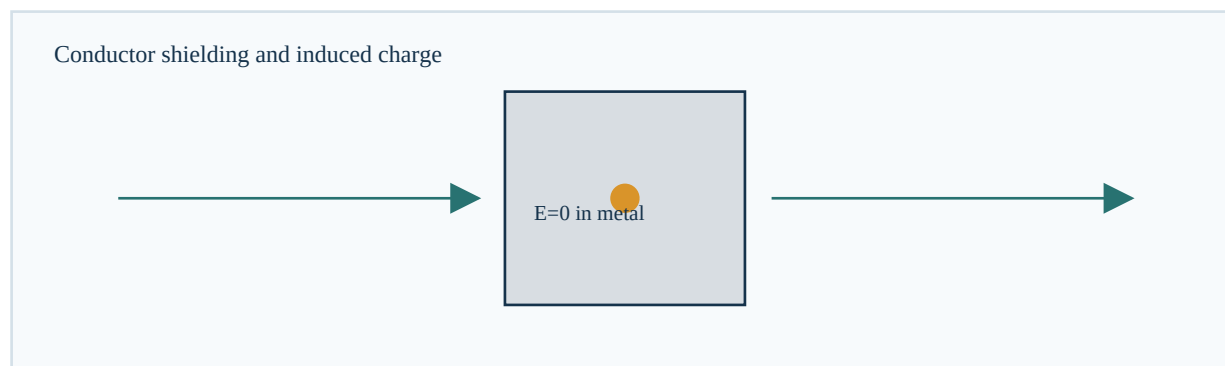
**Concept check:** Compare the field-distance dependence for an infinite line charge and a uniformly charged spherical shell outside the shell. Explain why their powers of  $r$  differ.

**Answer and explanation:** Doubling distance from an infinite line halves the field because cylindrical area grows in proportion to  $r$ . Inside a charged spherical shell, a Gaussian sphere encloses no net charge, so Gauss law gives  $E=0$ .

## 07. Conductors, shielding, and induced charge

**Scope and source status:** Electrostatic equilibrium, zero interior field, surface charge, cavities, shielding, pointed conductors, molecular induction.

**Question-generation tags:** draw-phenomenon explain-phenomenon fill-in



### Core understanding

- Free charges in a conductor move until the electric field inside the metal is zero. Any remaining tangential field would continue driving surface charge, so the equilibrium field is normal to the surface.
- Excess charge resides on surfaces. A charge in a cavity induces opposite charge on the inner wall; a neutral conductor then places compensating charge on its outer surface.
- A closed conductor shields its cavity from external static fields. Sharp points have high surface-charge density and therefore unusually large nearby fields.

#### Hallmark equation

$$E_{\text{metal}} = 0 \quad E_{\text{outside}} = \sigma / \epsilon_0$$

**Symbols:**  $E_{\text{metal}}$  is the field inside conducting material and  $\sigma$  is surface charge density. **Physical meaning:** free charges rearrange until no force remains in the metal; the normal surface field is the field produced by the accumulated surface charge just outside.

**Concept check:** Explain why a static electric field cannot remain inside conducting material, and describe the induced charge on the inner wall when a positive charge is placed in a neutral cavity.

**Answer and explanation:** A tangential field would exert a force along the conductor and keep free charges moving, contradicting electrostatic equilibrium. A  $+Q$  charge in a cavity induces  $-Q$  on the cavity wall so that the conducting material itself encloses zero net charge.

## 08. Electric potential, energy, and equipotentials

**Scope and source status:**  $V=U/q$ , work, potential differences, point and continuous sources, equipotential surfaces,  $E=-\text{grad } V$ .

**Question-generation tags:** fill-in diagram-matching qualitative-prediction derivation

Equipotential map and field direction



**E is perpendicular to equipotentials**

### Core understanding

- Potential  $V$  is scalar energy per charge, so point-charge potentials add algebraically. Potential differences are physical even though the zero of  $V$  is chosen by convention.
- The field does work when a charge moves through a potential difference:  $\Delta U = q \Delta V$ . Equipotential motion has  $\Delta V = 0$  and therefore zero electric work.
- $E = -\text{grad } V$  means the field points in the direction of steepest decrease of potential. Closely spaced equipotentials indicate a stronger field.

#### Hallmark equation

$$\mathbf{E} = -\text{grad } V \quad \Delta U = q \Delta V$$

**Symbols:**  $V$  is potential energy per charge,  $U$  is potential energy,  $q$  is the charge being moved, and  $\nabla V$  describes spatial change in potential. **Physical meaning:** the field points in the direction where a positive charge loses potential energy most rapidly; equipotential motion has no potential-energy change.

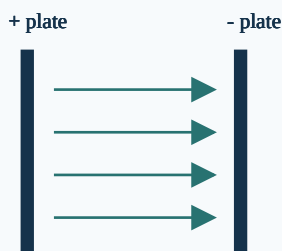
**Concept check:** Explain what it means physically for electric potential to decrease in a direction, and why the electric field is perpendicular to an equipotential surface.

**Answer and explanation:** If  $V$  decreases as  $x$  increases,  $dV/dx$  is negative and  $E_x = -dV/dx$  is positive. Motion along an equipotential has  $\Delta V = 0$ , so  $\Delta U = q \Delta V = 0$  and the electric field does no work.

## 09. Capacitance and dielectrics

**Scope and source status:** Capacitor geometry, series/parallel combinations, energy, battery-connected versus isolated cases, polarization and permittivity.  
**Question-generation tags:** matching fill-in qualitative-prediction

Parallel-plate capacitor



### Core understanding

- Capacitance is the charge stored per potential difference, determined by geometry and dielectric material. Parallel plates have  $C = \epsilon A/d$  when edge fringing is negligible.
- Parallel capacitors share voltage and add capacitance; series capacitors share charge magnitude and add reciprocals. The battery-connected or isolated condition determines what changes when a dielectric is inserted.
- A dielectric polarizes: bound charges reduce the internal field for fixed free charge and allow more charge to be stored at the same voltage. Stored energy is energy in the electric field.

#### Hallmark equation

$$C = Q/V \quad U = \frac{1}{2} C V^2$$

**Symbols:** C is capacitance, Q is stored free charge, V is the potential difference, and U is stored field energy. **Physical meaning:** geometry and material determine how much charge a capacitor can hold for a given voltage; a dielectric polarizes and changes that charge-voltage relationship.

**Concept check:** Explain what capacitance measures, how parallel and series connections differ physically, and why inserting a dielectric changes the stored charge or voltage depending on whether a battery remains connected.

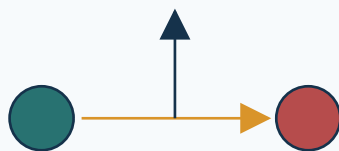
**Answer and explanation:** In parallel, every capacitor has the same voltage and the stored charges add, so capacitances add. A dielectric polarizes: bound charge reduces the internal field for fixed free charge and increases the capacitance.

## 10. Dipoles, torque, polarization, and displacement

**Scope and source status:** Dipole moment direction, torque, potential energy, polar/nonpolar molecules, P, D, susceptibility, free versus bound charge.

**Question-generation tags:** diagram-matching explain-phenomenon derivation

Dipole torque and polarization



torque rotates p toward E

### Core understanding

- A dipole has equal and opposite charges separated by  $d$ , with  $p=qd$  directed from negative to positive. In a uniform field, opposite forces can cancel while their separated lines of action create torque.
- Torque aligns  $p$  with  $E$ , and  $U=-p \cdot E$  makes parallel alignment stable. In a nonuniform field, a dipole can also translate toward stronger field.
- Polarization  $P$  is dipole moment per volume.  $D=\epsilon_0 E+P$  separates the free-charge bookkeeping from the material response;  $P$  and bound charge are not additional free charge.

#### Hallmark equation

$$\mathbf{p} = q \mathbf{d} \quad \tau = p E \sin\theta \quad \mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P}$$

**Symbols:**  $p$  is dipole moment,  $d$  is charge separation,  $\tau$  is torque,  $E$  is external field,  $P$  is dipole moment per volume, and  $D$  combines free-charge field with polarization. **Physical meaning:** separated opposite forces form a turning couple, while microscopic dipole alignment produces a macroscopic material response.

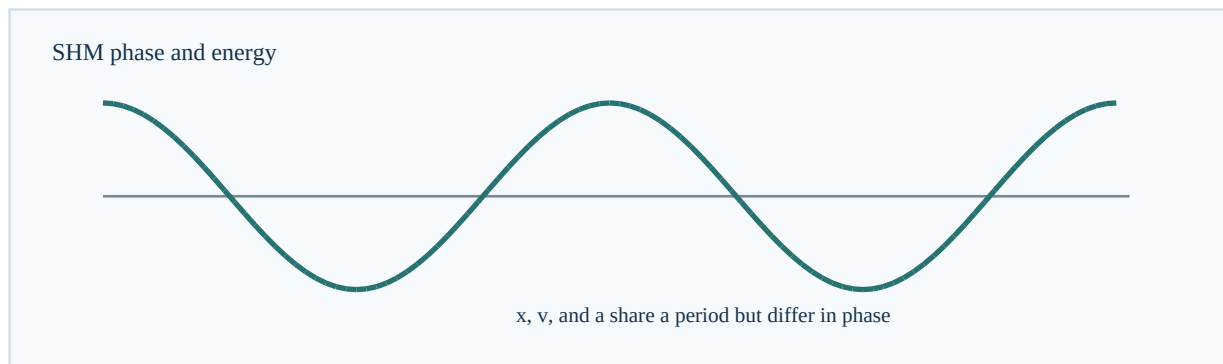
**Concept check:** Explain how a dipole can have zero net force but nonzero torque in a uniform electric field, and identify the stable orientation of its dipole moment.

**Answer and explanation:** The stable orientation is  $p$  parallel to  $E$  because  $U=-pE \cos(\theta)$  is then smallest. Polarization  $P$  is the net dipole moment per unit volume, not the total free charge density.

## 11. Oscillations and simple harmonic motion

**Scope and source status:** Hooke law, SHM differential equation, amplitude, phase, period,  $x/v/a$  phase, energy, phasors, resonance background.

**Question-generation tags:** graph-interpretation draw-phenomenon derivation



### Core understanding

- SHM requires a restoring acceleration proportional to displacement:  $x \text{ double-dot} = -\omega^2 x$ . A mass-spring system has  $\omega = \sqrt{k/m}$ , so stiffness speeds oscillation and mass slows it.
- The phase of  $x$ ,  $v$ , and  $a$  matters:  $v$  is a quarter-cycle shifted from  $x$ , while  $a$  is opposite  $x$ . Turning points have zero speed and maximum restoring acceleration.
- Energy moves between kinetic and spring potential while the total remains constant in ideal SHM. Damping removes energy; driving near the natural frequency produces resonance.

#### Hallmark equation

$$x = A \cos(\omega t + \phi) \quad \omega = \sqrt{k/m}$$

**Symbols:**  $A$  is amplitude,  $\omega$  is angular frequency,  $t$  is time,  $\phi$  is initial phase,  $k$  is spring stiffness, and  $m$  is mass.

**Physical meaning:** the sinusoid is the motion whose acceleration always points toward equilibrium and whose restoring strength grows with displacement.

**Concept check:** At the positive turning point of an ideal mass-spring oscillator, describe the velocity, acceleration, spring energy, and direction of the restoring force.

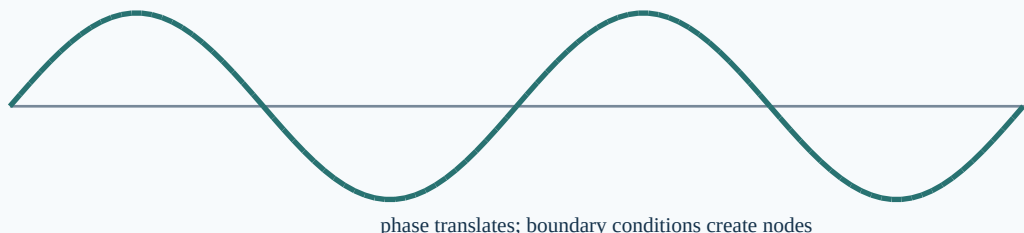
**Answer and explanation:** At maximum positive displacement, velocity is zero and acceleration is maximum negative, pointing toward equilibrium. Increasing mass increases the period because  $T = 2\pi \sqrt{m/k}$ .

## 12. Mechanical waves, interactions, and standing waves

**Scope and source status:** Pulses, traveling wave translation, transverse/longitudinal motion, wave speed, linear wave equation, reflection, transmission, interference, standing waves.

**Question-generation tags:** fill-in diagram-matching derivation draw-phenomenon

Traveling wave and standing-wave modes



### Core understanding

- A traveling wave carries energy while each medium element oscillates locally. In  $y=f(x-vt)$ , constant phase moves right; the pattern speed is not the same as the particle speed.
- The harmonic form identifies  $A$ ,  $k$ ,  $\omega$ , phase, wavelength, frequency, and direction. The linear wave equation relates spatial curvature to temporal acceleration when the medium has constant wave speed.
- Reflection depends on the boundary, interference adds displacements, and two opposite waves form standing nodes and antinodes. Fixed-end conditions select  $\lambda_n = 2L/n$ .

#### Hallmark equation

$$y = A \cos(kx - \omega t + \phi) \quad v = f\lambda$$

**Symbols:**  $A$  is displacement amplitude,  $k$  is wave number,  $\omega$  is angular frequency,  $\phi$  is phase,  $\lambda$  is wavelength,  $f$  is frequency, and  $v$  is waveform speed. **Physical meaning:** the phase argument tracks a repeating pattern in both space and time; the sign before  $\omega t$  selects travel direction and  $v=f\lambda$  links spatial and temporal repetition.

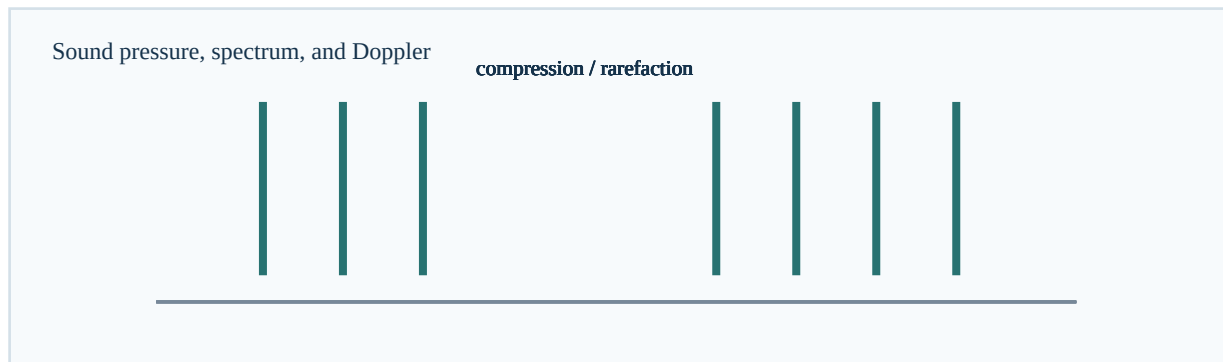
**Concept check:** For  $y=f(x-vt)$ , explain how you know the wave moves right, and distinguish the speed of the waveform from the velocity of a point in the medium.

**Answer and explanation:** The form  $f(x-vt)$  moves right because constant phase requires  $x$  to increase with  $t$ . At a boundary the source-set frequency remains continuous, while speed and wavelength adjust to the new medium.

## 13. Sound, intensity, spectra, hearing, and Doppler effect

**Scope and source status:** Sound production/detection, compression and rarefaction, pressure amplitude, intensity, decibels, harmonics, noise, hearing, Doppler shift.

**Question-generation tags:** fill-in graph-interpretation explain-phenomenon



### Core understanding

- Sound is a longitudinal pressure disturbance: compressions are high-pressure regions and rarefactions are low-pressure regions. Sound needs a material medium.
- Intensity is power per area and falls as  $1/r^2$  for an isotropic source. Decibels describe ratios logarithmically, so equal dB changes represent multiplicative intensity changes.
- Frequency sets pitch, amplitude and pressure relate to loudness, and harmonic content sets timbre. Relative motion changes wavefront arrival rate: approach raises perceived frequency and recession lowers it.

#### Hallmark equation

$$I = P/(4\pi r^2) \quad \beta = 10 \log_{10}(I/I_0)$$

**Symbols:**  $I$  is intensity,  $P$  is acoustic power,  $r$  is distance,  $\beta$  is sound level, and  $I_0$  is the reference intensity. **Physical meaning:** spherical spreading lowers intensity with distance, while the logarithm expresses how hearing compares multiplicative intensity ratios. Doppler shifts arise from arrival rate, not from a change in source power.

**Concept check:** An observer moves toward a stationary sound source. Predict the change in perceived frequency and explain it using wavefront arrival rate rather than source loudness.

**Answer and explanation:** Doubling the distance quarters intensity because the same power spreads over four times the spherical area. An approaching source or observer increases wavefront arrival rate, so perceived frequency rises.

## References and provenance

Primary: saved SingleFile Canvas pages in docs/reference/canvas\_pages and worksheet scans in docs/reference/reference-texts/Physics 2 scans Test 1. Supplemental: OpenStax University Physics Volume 1, Chapters 15-17; OpenStax University Physics Volume 2, Chapters 5-7; Young and Freedman, University Physics, relevant electrostatics, oscillations, waves, and sound chapters.

Original captures and scans were preserved unchanged. Repeated page titles were normalized in the knowledge base. Figures in this package are clean instructional redraws; the source inventory records where professor visuals are available for later extraction and reuse.

# 01 ELECTRIC FORCE, FIELDS, AND CONSERVATION

- Charge is a property of matter: electrons carry negative charge, protons positive charge, and charge is conserved in every isolated interaction.
- Coulomb force is central and inverse-square:  $F = k|q_1q_2|/r^2$ . Like charges repel; unlike charges attract along the line joining them.
- The electric field is the force a tiny positive test charge would feel:  $E = F/q$ . The field exists whether or not a test charge is present.
- For a positive source charge, E points away; for a negative source charge, E points toward. The force on a negative test charge is opposite E.
- Electric and gravitational fields share the inverse-square pattern, but gravity is always attractive while electric force can attract or repel.
- A field diagram is a map of direction and relative strength: lines never cross, crowding means larger magnitude, and arrows show the force on positive charge.
- Conservation-law thinking is a fast check: charge can move between objects, but the algebraic total before and after remains fixed.
- Units: force is newtons, charge coulombs, field N/C (equivalently V/m). Keep source charge and test charge roles distinct.

## VISUAL MAP



field direction from a positive source

## DERIVATION | FROM COULOMB FORCE TO FIELD

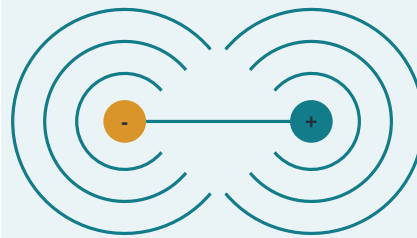
Start with  $F = kqQ/r^2$ . Divide by the test charge  $q$ :  $E = F/q = kQ/r^2$ . The test charge disappears, showing that E belongs to the source arrangement. Restore direction with  $\hat{r}$ , so  $E = kQ \hat{r}/r^2$ .

$$\text{Hallmark: } F = qE \quad \text{and} \quad F_{12} = k q_1 q_2 / r^2 \hat{r}$$

## 02 SUPERPOSITION, VECTORS, DISTRIBUTIONS, AND DIPOLES

- Fields add by superposition: calculate each source contribution at the same observation point, resolve into components, then add vectors—not magnitudes.
- Symmetry can cancel components. At the midpoint of equal like charges, horizontal components cancel; between opposite charges, the fields reinforce in one direction.
- For many point charges,  $E = \sum kq_i \hat{r}_i / r_i^2$ . Draw a local coordinate system before substituting numbers.
- A continuous distribution replaces the sum with an integral:  $dq = \lambda dx$  for a line,  $\sigma dA$  for a surface, or  $\rho dV$  for a volume.
- An electric dipole is two equal opposite charges separated by distance  $d$ . Its dipole moment  $p$  points from negative to positive charge.
- Far from a dipole, field strength falls faster than a single-charge field because opposite contributions nearly cancel; direction depends strongly on location.
- Field vectors point tangent to field lines. At a point, use the geometry of the source arrangement to anticipate zeroes, maxima, and limiting behavior.
- A reliable workflow is: sketch  $\rightarrow$  choose axes  $\rightarrow$  write each vector  $\rightarrow$  project  $\rightarrow$  sum  $\rightarrow$  check units and the limiting case  $r \rightarrow \infty$ .

### VISUAL MAP



dipole moment points from - to +

### DERIVATION | SUPERPOSITION AND A CONTINUOUS CHARGE

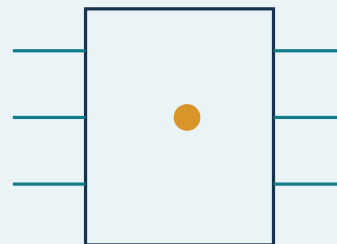
For separated pieces,  $E = \sum dE_i$ . Let the pieces become infinitesimal:  $dq \rightarrow dq$  and  $\Sigma \rightarrow \int$ . With  $dq = \lambda dl$ ,  $\sigma dA$ , or  $\rho dV$ ,  $E = k \int (dq/r^2) \hat{r}$ . Geometry determines the component integrals.

**Dipole idea:  $p = qd$  ; far-field scale  $E \propto p/r^3$**

### 03 CONDUCTORS AND ELECTROSTATIC EQUILIBRIUM

- In electrostatic equilibrium, free charges in a conductor have stopped drifting. Therefore the electric field inside the conducting material is zero.
- If an internal field remained, mobile charges would accelerate. They rearrange on the surface until their own field cancels the applied field in the metal.
- The conductor is an equipotential: if two points had different  $V$ , tangential electric forces would drive surface charge along the conductor.
- At a surface,  $E$  has no tangential component and points perpendicular to the conductor. Just outside,  $E_{\perp} = \sigma/\epsilon_0$ .
- Charge placed on an isolated conductor resides on the outer surface unless a cavity contains charge. In a cavity, induced inner-surface charge balances the enclosed charge.
- Electrostatic shielding follows from the zero interior field: a conducting enclosure protects its interior from external static fields, provided the geometry is closed.
- Sharp points accumulate surface charge more densely, producing stronger nearby fields and enabling corona discharge or lightning-rod action.
- For conductor questions, identify the material region, cavity, and exterior first; apply Gauss's law only after the equilibrium constraints are stated.

#### VISUAL MAP



$E = 0$  inside metal; surface charge shields

#### DERIVATION | WHY A CONDUCTOR HAS $E = 0$ INSIDE

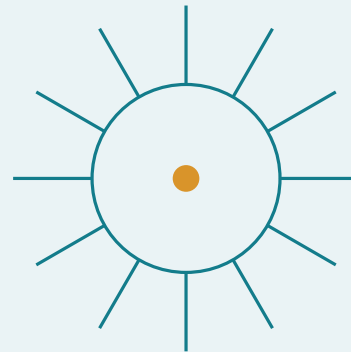
Assume a nonzero field in the metal. Free charge would feel  $F = qE$  and keep moving, contradicting equilibrium. Charge therefore rearranges until the interior field cancels. A Gaussian surface wholly inside the metal then has  $\oint \mathbf{E} \cdot d\mathbf{A} = 0$ , so  $Q_{\text{enc}} = 0$ .

$$\text{Surface condition: } E_{\text{inside metal}} = 0 \quad ; \quad E_{\text{just outside}} = \sigma/\epsilon_0$$

## 04 GAUSS'S LAW AND GAUSSIAN-SURFACE SYMMETRY

- Electric flux measures how much field passes through a surface:  $\Phi_E = \int \mathbf{E} \cdot d\mathbf{A}$ . The area vector points outward for a closed surface.
- Gauss's law says total outward flux equals enclosed charge divided by  $\epsilon_0$ . Charges outside the surface can alter local field but contribute zero net flux.
- The law is always true, but it becomes a shortcut only when symmetry makes  $E$  constant in magnitude and direction over useful parts of the surface.
- Choose a Gaussian sphere for spherical symmetry, cylinder for an infinite line, and pillbox for an infinite plane or conductor surface.
- The surface is imaginary; it is chosen to match the field geometry, not to follow the physical boundary of the charged object.
- On a surface where  $E$  is constant and perpendicular,  $\Phi = EA$ . Where  $E$  is tangent,  $\mathbf{E} \cdot d\mathbf{A} = 0$ . Where  $E$  varies, integrate or use smaller elements.
- Flux is signed: outward field gives positive contribution, inward field negative. Zero net flux does not mean zero field everywhere.
- The key physical thought is “count field lines in aggregate”: enclosed charge is the source of net field-line excess through a closed boundary.

### VISUAL MAP



Gaussian surface: choose symmetry first

### DERIVATION | GAUSS LAW FROM A SPHERICAL FIELD

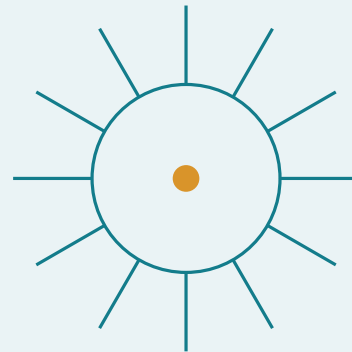
For a point charge,  $E = kQ/r^2$  and  $dA$  is radial, so  $\int \mathbf{E} \cdot d\mathbf{A} = E \int dA = (kQ/r^2)(4\pi r^2) = 4\pi kQ = Q/\epsilon_0$  because  $k = 1/(4\pi\epsilon_0)$ . The same flux result extends to any closed surface.

$$\text{Gauss: } \int \mathbf{E} \cdot d\mathbf{A} = Q_{\text{enc}} / \epsilon_0$$

## 05 APPLICATIONS OF GAUSS'S LAW

- For an infinite line with linear density  $\lambda$ , use a coaxial cylinder of radius  $r$  and length  $L$ . Only the curved side contributes:  $\Phi = E(2\pi rL)$ .
- The enclosed charge is  $\lambda L$ , so  $E = \lambda/(2\pi\epsilon_0 r)$ . The  $1/r$  dependence reflects cylindrical spreading rather than spherical spreading.
- For a spherical shell or point-like spherical distribution, use a Gaussian sphere. Outside, the field is the same as if all net charge were concentrated at the center.
- Inside a uniformly charged solid sphere,  $Q_{\text{enc}}$  grows with  $r^3$  while area grows with  $r^2$ , giving  $E \propto r$ .
- For a nonuniform sphere, compute  $Q_{\text{enc}}(r) = \int \rho(r')dV$  before applying Gauss's law. Do not substitute total charge inside the interior surface.
- A conducting shell has  $E = 0$  in the metal. A Gaussian surface in the metal then forces the enclosed algebraic charge to be zero.
- Outside a charged conductor, spherical symmetry is required before replacing the object with a point charge; arbitrary shapes need not have a radial field.
- Always state the region—inside, within metal, cavity, or outside—because the enclosed charge and symmetry can change at each boundary.

### VISUAL MAP



Gaussian surface: choose symmetry first

### DERIVATION | UNIFORM SOLID SPHERE, INSIDE AND OUTSIDE

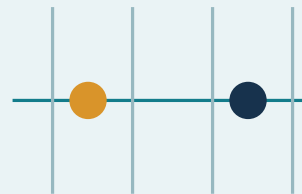
Inside radius  $r$ ,  $Q_{\text{enc}} = \int_0^r \rho(4\pi r'^2)dr' = 4\pi\rho r^3/3$ . Gauss gives  $E(4\pi r^2) = Q_{\text{enc}}/\epsilon_0$ , hence  $E = \rho r/(3\epsilon_0)$ . Outside, replace  $r$  by  $R$  in  $Q_{\text{total}}$ , giving  $E = kQ_{\text{total}}/r^2$ .

**Line charge result:  $E(r) = \lambda/(2\pi\epsilon_0 r)$  ; solid sphere:  $E = \rho r/(3\epsilon_0)$**

## 06 ELECTRIC POTENTIAL AND ENERGY

- Potential  $V$  is electric potential energy per unit charge. It is a scalar, so contributions from multiple charges add algebraically instead of vectorially.
- A potential difference measures work per charge:  $dV = -\int \mathbf{E} \cdot d\mathbf{l}$ . A positive charge naturally moves toward lower potential when released.
- For a point charge,  $V = kq/r$  with  $V = 0$  chosen at infinity. Positive  $q$  produces positive  $V$ ; negative  $q$  produces negative  $V$ .
- Electric potential energy is  $U = qV$ . The sign of  $q$  matters: a negative charge can lose  $U$  while moving toward higher  $V$ .
- The electric field points in the direction of decreasing potential. In one dimension,  $E_x = -dV/dx$ ; a steep potential graph means a strong field.
- Equipotential surfaces have constant  $V$ , so moving along one requires no electric work and  $\mathbf{E}$  is perpendicular to the surface.
- Use potential for energy and geometry problems; use field for force and acceleration. They describe the same interaction from complementary viewpoints.
- Checks: voltage is joules per coulomb, field is volts per metre, and potential must become nearly uniform far from a localized charge distribution.

### VISUAL MAP



equipotential lines;  $\mathbf{E}$  points downhill in  $V$

### DERIVATION | WHY $\mathbf{E} = -\nabla V$

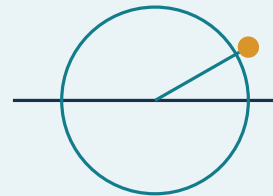
Work by the field is  $dW = \mathbf{F} \cdot d\mathbf{l} = q\mathbf{E} \cdot d\mathbf{l}$ . Since  $dU = -dW$  and  $dU = qdV$ ,  $qdV = -q\mathbf{E} \cdot d\mathbf{l}$ . Cancel  $q$ :  $dV = -\mathbf{E} \cdot d\mathbf{l}$ . Component-by-component,  $E_x = -dV/dx$ , etc., which combine as  $\mathbf{E} = -\nabla V$ .

**Field–potential link:  $\mathbf{E} = -\nabla V$  ;  $dU = qdV$**

## 07 SIMPLE HARMONIC MOTION

- Simple harmonic motion occurs when the restoring force is proportional to displacement and opposite in direction:  $F = -kx$ .
- For a mass-spring oscillator, Newton's law gives  $mx'' = -kx$ , so  $\omega = \sqrt{k/m}$ ,  $T = 2\pi/\omega$ , and  $f = 1/T$ .
- The amplitude  $A$  is the maximum displacement; it sets the energy but not the period in ideal linear SHM. Phase tells where the cycle begins.
- A convenient model is  $x(t) = A \cos(\omega t + \phi)$ . Then  $v$  is shifted by a quarter cycle and acceleration is opposite  $x$ .
- At turning points,  $x = \pm A$ , speed is zero, and spring energy is maximum. At equilibrium,  $x = 0$ , speed is maximum, and kinetic energy is maximum.
- The graphs of  $x$ ,  $v$ , and  $a$  are sinusoids with the same period:  $v$  leads  $x$  by  $90^\circ$ , while  $a$  is  $180^\circ$  out of phase with  $x$ .
- A phasor is a rotating-radius picture whose horizontal projection gives  $x(t)$ . It turns phase comparisons into geometry.
- Energy conservation provides a robust check:  $E = \frac{1}{2}kx^2 + \frac{1}{2}mv^2 = \frac{1}{2}kA^2$  for an undamped oscillator.

### VISUAL MAP



phasor projection gives  $x(t)$

### DERIVATION | MASS-SPRING SHM

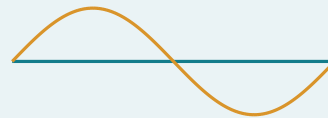
Hooke's law gives  $F = -kx$ . Newton's second law gives  $mx'' = -kx$ . Divide by  $m$  and define  $\omega^2 = k/m$ :  $x'' + \omega^2 x = 0$ . A cosine satisfies this because  $d^2[\cos(\omega t)]/dt^2 = -\omega^2 \cos(\omega t)$ .

$$\text{SHM: } x'' + \omega^2 x = 0 \quad ; \quad \omega = \sqrt{k/m} \quad ; \quad E = \frac{1}{2}kA^2$$

## 08 TRAVELING WAVES AND THE LINEAR WAVE EQUATION

- A traveling wave is a pattern that moves while the medium's particles oscillate locally. Energy travels with the disturbance; matter usually oscillates about equilibrium.
- A right-moving shape is written  $y(x,t) = f(x - vt)$ ; the minus sign makes a fixed feature occur at larger  $x$  as time increases.
- For a sinusoid,  $y = A \cos(kx - \omega t + \phi)$ , where  $k = 2\pi/\lambda$ ,  $\omega = 2\pi f$ , and  $v = \omega/k = f\lambda$ .
- Phase compares locations in the cycle. Points separated by  $\lambda$  are in phase; separation  $\lambda/2$  means opposite displacement for a sinusoid.
- Transverse waves oscillate perpendicular to travel; longitudinal waves oscillate parallel to travel, producing compressions and rarefactions.
- The linear wave equation follows from local restoring forces plus inertia: each small element accelerates according to the curvature of its neighbors.
- For a string, tension  $T$  and linear density  $\mu$  give  $v = \sqrt{T/\mu}$ . Greater tension speeds the wave; greater mass per length slows it.
- The linear approximation assumes small slopes and independent superposition. Large amplitudes, nonlinear media, or strong dispersion can break the simple model.

### VISUAL MAP



a shape translates without changing its form

### DERIVATION | LINEAR WAVE EQUATION

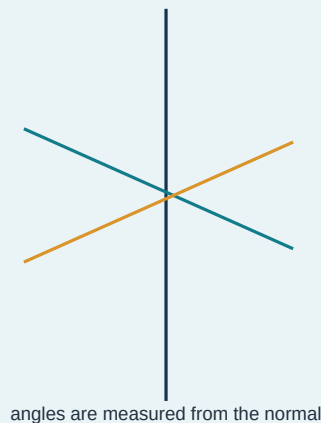
For  $y = A \cos(kx - \omega t)$ , differentiating twice gives  $y_{xx} = -k^2 y$  and  $y_{tt} = -\omega^2 y$ . Therefore  $y_{xx} = (k^2/\omega^2)y_{tt} = (1/v^2)y_{tt}$  because  $v = \omega/k$ . The equation encodes equal propagation speed in both directions.

$$\text{Wave equation: } d^2y/dx^2 = (1/v^2) d^2y/dt^2 \quad ; \quad v = f\lambda$$

## 09 REFLECTION, REFRACTION, AND SNELL'S LAW

- Reflection reverses the direction component normal to a boundary while preserving the tangential component. The angle of incidence equals the angle of reflection.
- Refraction occurs when wave speed changes across an interface. Frequency is fixed by the source, so wavelength changes according to  $\lambda = v/f$ .
- Huygens' principle treats every point on a wavefront as a source of secondary wavelets. Their envelope predicts the next wavefront and its changed direction.
- At an interface, phase must match along the boundary. The tangential wavelength component is therefore continuous, leading to  $n_1 \sin \theta_1 = n_2 \sin \theta_2$ .
- The refractive index is  $n = c/v$ . Entering a slower medium bends the ray toward the normal; entering a faster medium bends it away.
- A derivation strategy is geometric: compare equal travel times along the interface, substitute  $v = c/n$ , and rearrange the sine ratio.
- Reflection and refraction can coexist. The amount of transmitted versus reflected energy depends on the boundary and wave properties.
- Watch the angle convention:  $\theta$  is measured from the normal, not from the surface. Frequency does not jump at the boundary.

### VISUAL MAP



### DERIVATION | SNELL LAW FROM PHASE MATCHING

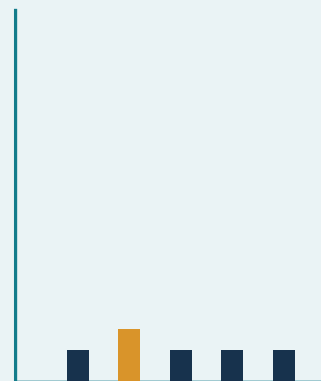
During the same time interval, wavefronts advance  $v_1 dt$  and  $v_2 dt$ . Matching phase along the boundary makes the tangential distances equal:  $v_1 dt \sin \theta_1 = v_2 dt \sin \theta_2$ . Cancel  $dt$  and use  $n = c/v$ :  $\sin \theta_1 / \sin \theta_2 = v_2 / v_1 = n_1 / n_2$ , so  $n_1 \sin \theta_1 = n_2 \sin \theta_2$ .

$$\text{Snell: } n_1 \sin \theta_1 = n_2 \sin \theta_2 \quad ; \quad f_1 = f_2$$

## 10 SOUND, NOISE, PRESSURE LEVEL, AND SPECTRUM

- Sound is a mechanical disturbance of matter. In air it is primarily longitudinal: oscillating molecules create alternating compressions and rarefactions.
- A pure tone has one dominant frequency; noise contains many frequencies with irregular phase and amplitude. A spectrum displays amplitude versus frequency.
- Sound pressure variation  $dp$  is the pressure oscillation around atmospheric pressure. Larger pressure amplitude generally means greater perceived loudness.
- Intensity is average power per area and, for a sinusoidal sound, scales with pressure amplitude squared. Spherical spreading reduces intensity with distance.
- Sound-pressure level uses a logarithm:  $\beta = 10 \log_{10}(I/I_0)$  dB. A tenfold intensity ratio changes level by 10 dB.
- The reference intensity  $I_0$  is a conventional threshold, so decibels compare ratios rather than absolute linear amounts.
- Harmonics are integer multiples of a fundamental frequency. Their relative amplitudes help distinguish instruments and sources with the same pitch.
- Frequency determines pitch, amplitude relates to loudness, and waveform/spectrum determines timbre. Do not confuse frequency with propagation speed.
- For a wave in a medium, sound speed depends on the medium's stiffness and inertia; temperature and material properties matter more than source loudness.

### VISUAL MAP



spectrum: amplitude versus frequency

### DERIVATION | SOUND LEVEL AND INTENSITY

Intensity is power per area:  $I = P/A$ . A spherical wave spreads over  $A = 4\pi r^2$ , so  $I_2/I_1 = r_1^2/r_2^2$ . Define level relative to  $I_0$  by  $\beta = 10 \log_{10}(I/I_0)$ . A tenfold intensity ratio therefore adds 10 dB.

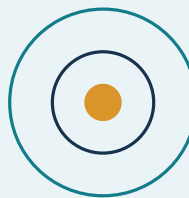
$$\text{Level: } \beta = 10 \log_{10}(I/I_0) \quad ; \quad I \propto (dp)^2$$

## 11 ATOMIC MODELS, IONS, AND MOLECULES

- Bohr's model places a compact nucleus at the center and treats electrons as occupying allowed energy states rather than arbitrary radii. It is a useful bridge between classical force ideas and quantization.
- For hydrogen, the inward electric attraction supplies centripetal force:  $k \frac{e^2}{r^2} = m_e v^2/r$ . This balance predicts the speed needed for circular motion at a chosen radius.
- The quantum model replaces definite electron orbits with an electron cloud: a probability map for where an electron is likely to be found. Darker or denser regions indicate greater probability, not a solid smear of charge.
- Ionization means supplying enough energy to remove a bound electron. The atom becomes a positive ion because it has more protons than electrons; the escaped electron is free.
- Atoms tend to gain or lose valence-shell electrons to reach a more stable outer configuration. Alkali metals commonly form  $\text{Na}^+$  or  $\text{K}^+$ ; chlorine commonly forms  $\text{Cl}^-$ .
- A molecule is a bonded collection of atoms:  $\text{H}_2$ ,  $\text{N}_2$ , and  $\text{CO}$  are diatomic;  $\text{H}_2\text{O}$  and  $\text{CO}_2$  are triatomic. The formula counts atoms, not net charge.
- In multi-electron atoms, each electron feels attraction toward the positive nucleus and repulsion from the other electrons. The net force is a vector sum, so simple one-electron circular pictures are only approximations.

### VISUAL MAP

cloud / allowed levels



nucleus

### DERIVATION | BOHR BALANCE AND IONIZATION

For a one-electron atom, set the electric attraction equal to the required centripetal force:  $ke^2/r^2 = m_e v^2/r$ . This does not by itself create the quantum model; quantization restricts which states are allowed. Ionization then means adding enough energy to move the electron from a bound state to an unbound state, leaving a positive ion.

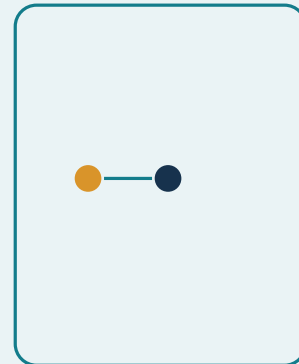
**Bohr force balance:**  $k \frac{e^2}{r^2} = m_e v^2/r$  ; **ionization:** bound electron + energy  $\rightarrow$  free electron + positive ion

## 12 DIELECTRICS, POLARIZATION, AND DISPLACEMENT

- A microscopic electric dipole consists of equal and opposite charges separated by vector  $d$ . Its dipole moment is  $p = qd$ , directed from negative charge to positive charge.
- A dielectric is an insulating material whose molecules can shift or rotate slightly in an applied field. The material does not conduct freely, but its internal charge separation changes the macroscopic field.
- Polarization  $P$  is the net dipole moment per volume:  $P = (1/V)\sum p_i$ . Random dipoles can give nearly zero  $P$ ; alignment produces a nonzero vector.
- The displacement field separates the response of free charge from material polarization:  $D = \epsilon_0 E + P$ . In free space  $P = 0$ , so  $D = \epsilon_0 E$ .
- For a linear, isotropic dielectric, polarization is proportional to the applied field:  $P = \epsilon_0 \chi_e E$ . Susceptibility  $\chi_e$  is dimensionless and measures how easily the material polarizes.
- Substitution gives  $D = \epsilon_0(1 + \chi_e)E = \epsilon E$ , where  $\epsilon$  is the material permittivity. This constitutive relation is a model assumption, not a universal law for every material or field strength.
- Keep free charge, bound charge,  $E$ ,  $P$ , and  $D$  conceptually distinct.  $E$  is the physical electric field;  $P$  describes dipole response;  $D$  is a bookkeeping field especially useful with free charge.

### VISUAL MAP

$E, P, D \rightarrow$



aligned dipoles

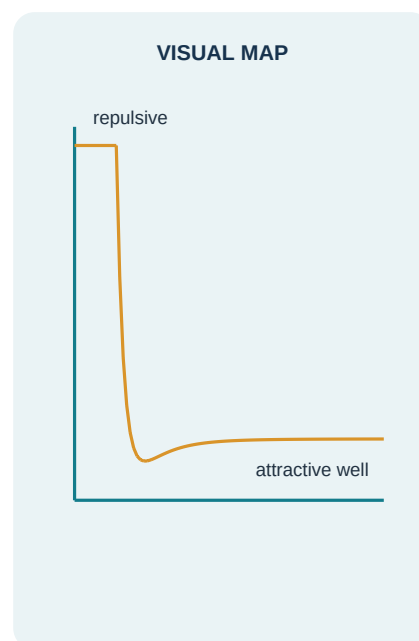
### DERIVATION | FROM DIPOLES TO $D = \epsilon E$

Start with one dipole  $p = qd$ . Sum all microscopic dipoles in volume  $V$  and divide by  $V$  to obtain  $P = (1/V)\sum p_i$ . Define  $D = \epsilon_0 E + P$ . For a linear isotropic material insert  $P = \epsilon_0 \chi_e E$ , factor  $\epsilon_0 E$ , and define  $\epsilon = \epsilon_0(1 + \chi_e)$ , giving  $D = \epsilon E$ .

$$\text{Dielectric chain: } p = qd \quad ; \quad P = (1/V)\sum p_i \quad ; \quad D = \epsilon_0 E + P = \epsilon E$$

## 13 POINT-CHARGE, IONIC, AND MOLECULAR POTENTIALS

- Electric potential energy tells whether a configuration is energetically favorable. For two point charges,  $U(r) = kq_1q_2/r$ , with zero chosen at infinite separation.
- Like charges give positive  $U$  and resist being brought together; opposite charges give negative  $U$  and lower their energy as they approach. The sign is a statement about energy, not a separate force direction rule.
- Force is the negative slope of the potential-energy curve:  $F(r) = -dU/dr$ . A steep slope means a large force; at a minimum, the slope is zero and the configuration is in equilibrium.
- Real ions such as  $\text{Na}^+$  and  $\text{Cl}^-$  are composite particles containing nuclei and electrons. Treating them as point charges is a useful far-field approximation, not a literal description at arbitrarily small  $r$ .
- The Lennard–Jones form  $U = C_{12}/r^{12} - C_6/r^6$  models short-range repulsion and longer-range attraction between neutral particles.
- At very small separation the repulsive term dominates; at intermediate separation the attractive term can create a potential well; far away both terms vanish and  $U$  approaches zero.
- For ionic solids or salts, a Coulomb term plus a steep repulsive term gives a Mie-type model. The exact constants depend on the interacting species and chosen approximation.



### DERIVATION | READING AN INTERACTION-POTENTIAL GRAPH

Plot  $U$  against separation  $r$ . The point-charge term varies as  $1/r$ ; additional short-range physics can create a steep positive wall and an attractive negative well. Differentiate the curve conceptually: negative slope means positive radial force under the chosen outward coordinate, positive slope means inward force, and a horizontal minimum gives  $F = 0$ .

**Potential–force link:**  $F(r) = -dU/dr$  ;  $U_{LJ} = C_{12}/r^{12} - C_6/r^6$